

GED Math Assignment 4

Part 1:

1. (a) $y = 3x - 4$ ① $\rightarrow$ into ②
 $x + y = 8$ ②

$$x + (3x - 4) = 8$$

$$4x - 4 = 8$$

$$4x = 8 + 4$$

$$x = \frac{12}{4}$$

$$x = 3 \rightarrow \text{into } ②$$

$$3 + y = 8$$

$$y = 8 - 3$$

$$y = 5$$

(b) $x = -4y + 5$ ①
 $x + 2y = 7$ ②
or

$$\begin{array}{r} x + 4y = 5 \\ - (x + 2y = 7) \\ \hline 0 + 2y = -2 \end{array}$$

$$y = -\frac{2}{2}$$

$$y = -1 \rightarrow \text{into } ①$$

$$x = -4(-1) + 5$$

$$x = 9$$

(c) ① $x + 2y = 3$ ($\times 2$)
② $5x + 4y = 8$

$$\begin{array}{r} 2x + 4y = 6 \\ - (5x + 4y = 8) \\ \hline -3x + 0 = -2 \end{array}$$

$$x = \frac{2}{3} \rightarrow \text{into } ②$$

$$5\left(\frac{2}{3}\right) + 4y = 8$$

$$\frac{10}{3} + 4y = 8$$

$$4y = \frac{24}{3} - \frac{10}{3}$$

$$4y = \frac{14}{3}$$

$$y = \frac{\frac{14}{3}}{4}$$

$$y = \frac{14}{12}$$

$$y = \frac{7}{6}$$

(d) $6x + 5y = 7$ ①
 $x - y = 3$ ②
 $x = y + 3 \rightarrow \text{into } ①$

$$6(y + 3) + 5y = 7$$

$$6y + 18 + 5y = 7$$

$$11y = 7 - 18$$

$$11y = -11 \rightarrow y = -\frac{11}{11}$$

$$y = -1$$

$$\downarrow$$

$$\text{into } ②$$

$$x - (-1) = 3$$

$$x + 1 = 3$$

$$x = 3 - 1$$

$$x = 2$$

$$(e) \quad \begin{aligned} 2m + n &= 2 \quad (1) \times 2 \\ 3m - 2n &= 3 \quad (2) \end{aligned}$$

$$\begin{aligned} 4m + 2n &= 4 \\ + (3m - 2n &= 3) \end{aligned}$$

$$7m + 0 = 7$$

$$m = \frac{7}{7}$$

$$m = 1 \rightarrow \text{into } (1)$$

$$2(1) + n = 2$$

$$n = 2 - 2$$

$$n = 0$$

$$(f) \quad \begin{aligned} 3a + 2b &= 4 \quad (1) \\ 2a + b &= 6 \quad (2) \end{aligned}$$

$$b = 6 - 2a \rightarrow \text{into } (1)$$

$$3a + 2(6 - 2a) = 4$$

$$3a + 12 - 4a = 4$$

$$12 - a = 4$$

$$-a = 4 - 12$$

$$a = 8 \rightarrow \text{into } (2)$$

$$2(8) + b = 6$$

$$16 + b = 6$$

$$b = 6 - 16$$

$$b = -10$$

$$(g) \quad \begin{aligned} 2x + y &= 4 \quad (1) \\ + (4x - y &= 2) \quad (2) \end{aligned}$$

$$6x + 0 = 6$$

$$x = \frac{6}{6}$$

$$x = 1 \rightarrow \text{into } (1)$$

$$2(1) + y = 4$$

$$y = 4 - 2$$

$$y = 2$$

$$(h) \quad \begin{aligned} 4x + 2y &= 7 \quad (1) \\ -x - 7y &= 6 \quad (2) \end{aligned}$$

$$-x = 6 + 7y$$

$$x = -6 - 7y \rightarrow \text{into } (1)$$

$$4(-6 - 7y) + 2y = 7$$

$$-24 - 28y + 2y = 7$$

$$-26y = 7 + 24$$

$$-26y = 31$$

$$y = -\frac{31}{26} \rightarrow \text{into } (2)$$

$$-x - 7\left(-\frac{31}{26}\right) = 6$$

$$-x + \frac{217}{26} = 6$$

$$-x = 6 - \frac{217}{26}$$

$$-x = \frac{156 - 217}{26}$$

$$x = \frac{217 - 156}{26}$$

$$x = \frac{61}{26}$$

3. Let S be the number of hours worked by Samantha and A be those worked by Adriana.

(a) $S = 2A$ ①

(b) $S + A = 39$ ②

(c) $2A + A = 39$ ②

$$3A = 39$$

$$A = \frac{39}{3}$$

$$A = 13 \rightarrow \text{into } ①$$

$$S = 2(13)$$

$$S = 26$$

$\therefore$ Adriana worked
13 hours and
Samantha 26.

4. (a) Let A be the number of goals and G be the number of assists. Let P the number of points awarded.

$$P = 2G + A$$
 ① and $P = 86$

$$G = A - 17$$
 ②

$$\begin{array}{r} 86 = 2G + A \\ + (-17 = G - A) \\ \hline \end{array}$$

$$69 = 3G + 0$$

$$G = \frac{69}{3}$$

$$G = 23 \rightarrow \text{into } ②$$

$$23 = A - 17$$

$$A = 23 + 17 \rightarrow A = 40$$

$\therefore$ Ugo had 40 assists
and 23 goals.

5. (a) Let m be the number of meals required
and C the total cost

$$C = 500 + 15m \quad (1)$$

$$C = 350 + 18m \quad (2)$$

$$(b) \quad \begin{array}{r} C = 500 + 15m \\ - (C = 350 + 18m) \\ \hline \end{array}$$

$$0 = 150 - 3m$$

$$3m = 150$$

$$m = \frac{150}{3}$$

$$m = 50 \rightarrow \text{int. } (1)$$

$$C = 500 + 15(50)$$

$$C = \$1250$$

$\therefore$ The number of
guests will be
50.

Part 2:

$$1. \quad -12 > x - 7$$

$$x < -12 + 7$$

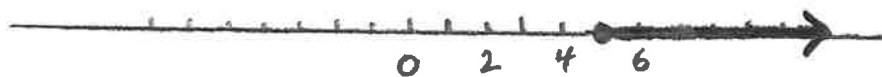
$$x < -5$$



$$2. \quad -1 + r \geq 4$$

$$r \geq 4 + 1$$

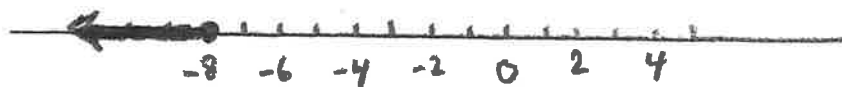
$$r \geq 5$$



$$3. \quad n - 6 \leq -14$$

$$n \leq -14 + 6$$

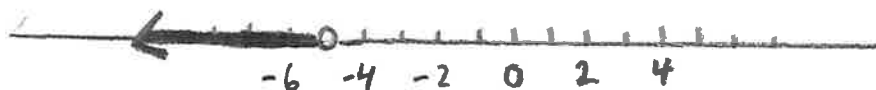
$$n \leq -8$$



$$4. \quad b - 7 < -12$$

$$b < -12 + 7$$

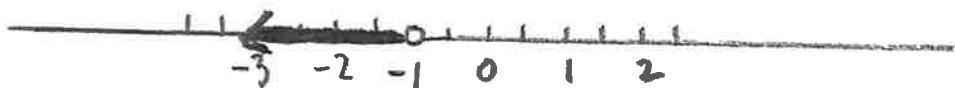
$$b < -5$$



$$9. \quad -3x > 3$$

$$x < \frac{3}{-3}$$

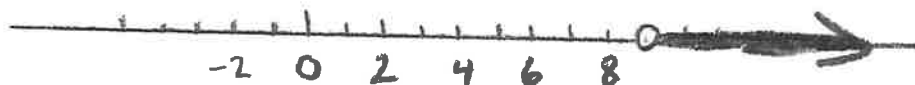
$$x < -1$$



$$10. \quad \frac{n}{3} > 3$$

$$n > (3)(3)$$

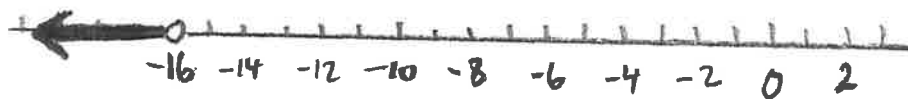
$$n > 9$$



$$11. \quad \frac{k}{4} < -4$$

$$k < (-4)(4)$$

$$k < -16$$



$$12. \quad -9x \geq -90$$

$$x \leq \frac{-90}{-9}$$

$$x \leq 10$$



$$19. -2(3+4n)-3 < 11+2n$$

$$-6-8n-3 < 11+2n$$

$$-8n-2n < 11+3+6$$

$$-10n < 20$$

$$n > \frac{-20}{10}$$

$$n > -2$$



$$20. 2(2-2v) > -10+3v$$

$$4-4v > -10+3v$$

$$-4v-3v > -10-4$$

$$-7v > -14$$

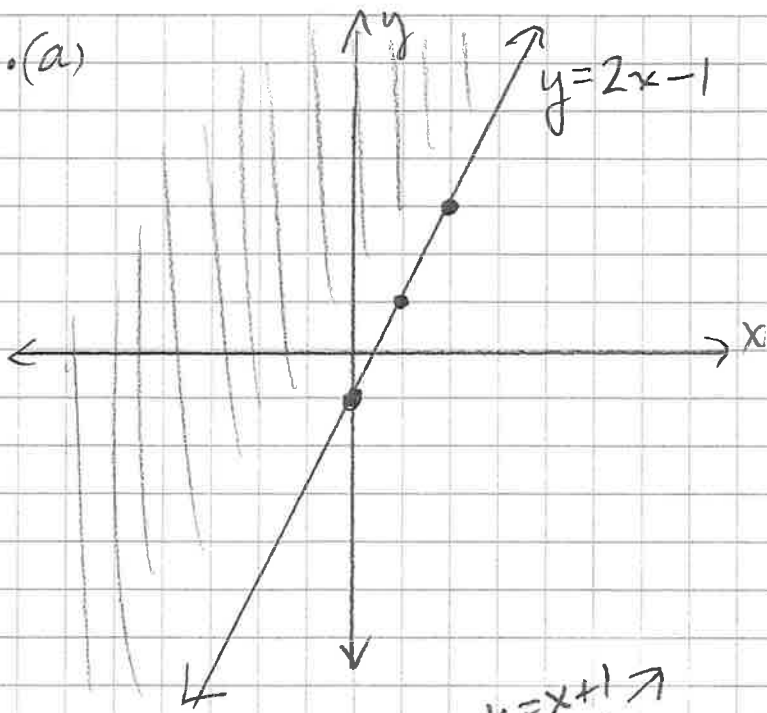
$$v < \frac{-14}{-7}$$

$$v < 2$$

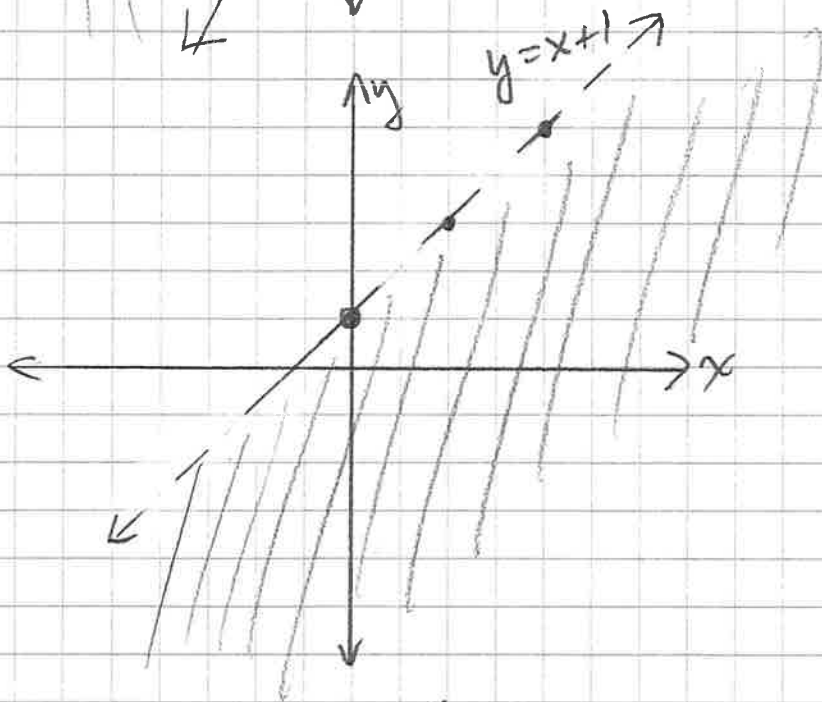


21. See graph paper.

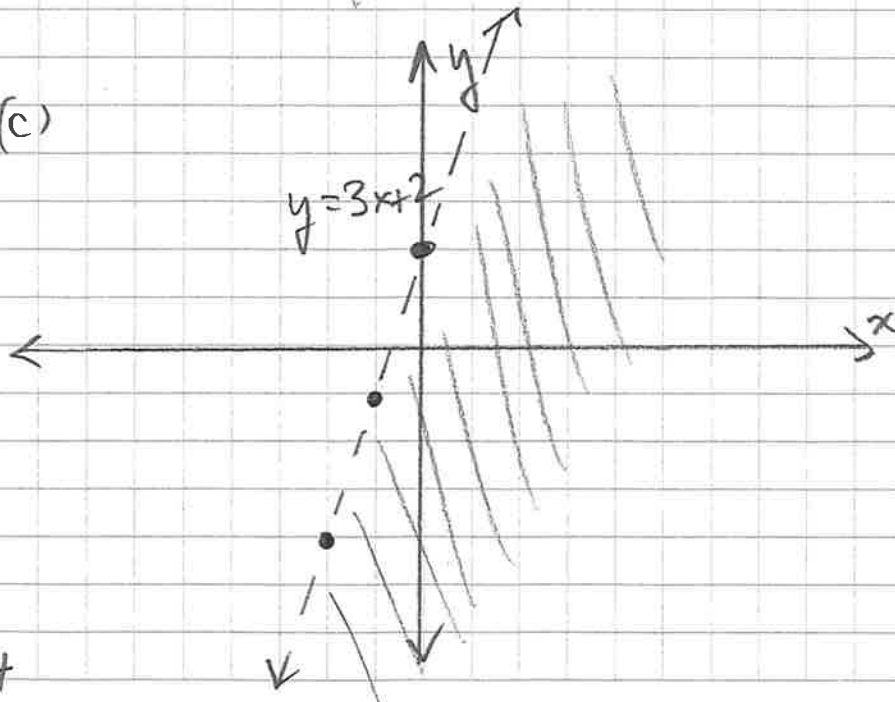
21.(a)



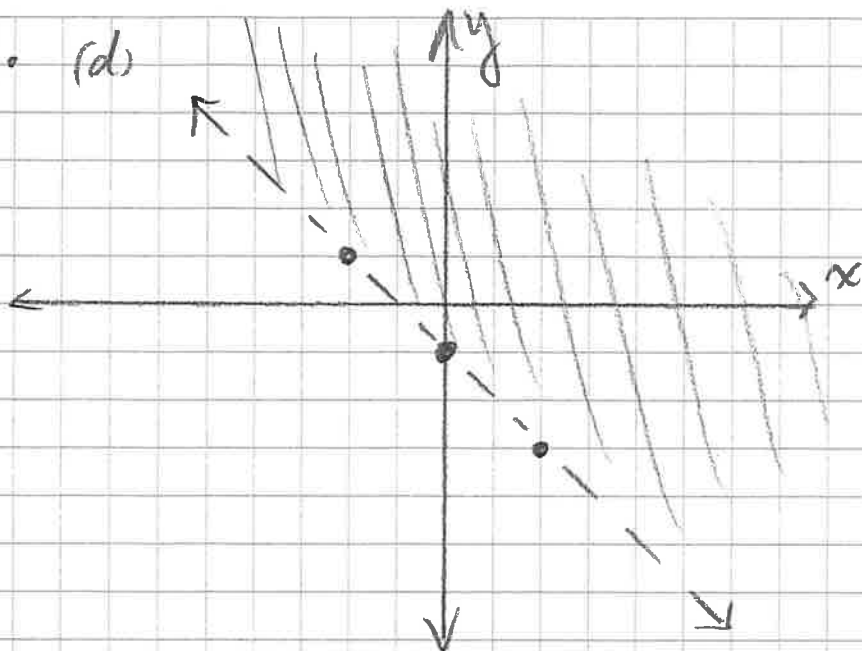
(b)



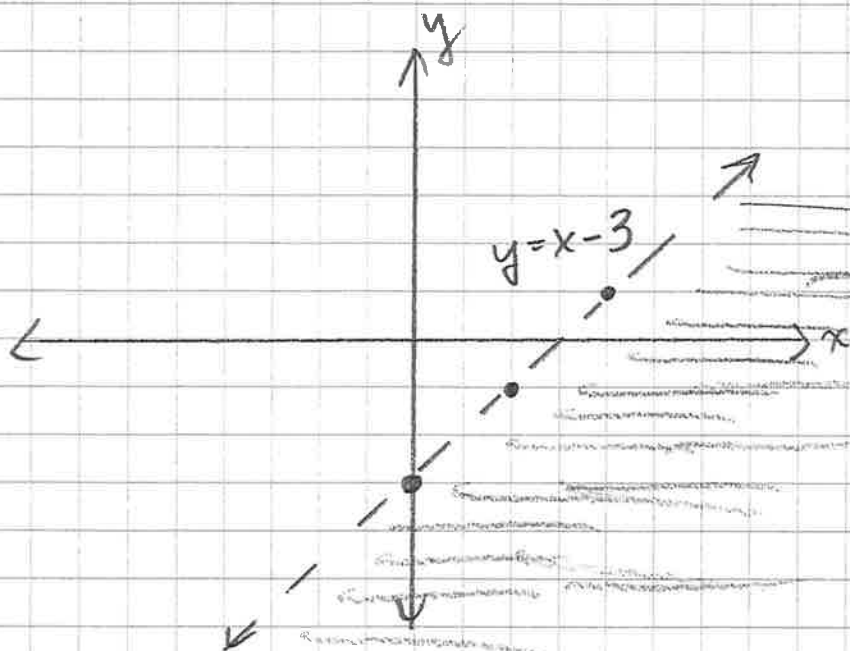
(c)



21. (d)



(e)



$$\begin{aligned} x - y &> 3 \\ -y &> 3 - x \\ y &< x - 3 \end{aligned}$$

(f)

